4

Inference in First-Order Predicate Logic

Chapter Learning Outcomes

CLO

4

- Apply unification algorithm.
- Convert FOPL sentence into conjunctive normal form (CNF).
- Skolemize logical expressions.
- Apply resolution algorithm.
- Solve some problems using resolution algorithm.

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Inference in FOPL

Resolution

- Unification
- Skolemization
- CNF

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Unification

Unify algorithm: takes 2 sentences p and q and returns a unifier if one exists.

EXAMPLE

Let p = knows(Ahmad, x)

q = knows(Ahmad, Adel).

Unify(p, q)

σ = {x/Adel}

Unification Algorithm

```
procedure unify(p, q, theta = [])
    Scan p and q left-to-right.
    If p and q are equal, return theta (success)
    Otherwise, let r and s be the terms in p and q,
      respectively, where disagreement first occurs
    If r is a variable that does not occur in s, then
      assign theta = theta + [r/s] and call
      unify(p[theta], q[theta], theta)
    else if s is a variable that does not occur in r, then
      assign theta = theta + [s/r] and call
      unify(p[theta], q[theta], theta))
    else return "failure"
```

Unification

EXAMPLE

Let p = knows(Ahmad, x)

q = knows(y, Adel).

Unify(p, q)

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Unification

EXAMPLE

Let p = knows(Ahmad, x)

q = knows(y, mother(y)).

Unify(p, q)

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Unification

EXAMPLE

Let p = knows(Ahmad, x)

q = knows(x, Adel).

Unify(p, q)

Quiz

1. Unify p(A, x) and p(A, B)

2. Unify p(A, x) and p(y, B)

3. Unify p(A, x) and p(y, f(y))

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Quiz

4. Unify p(A, x) and p(x, B)

5. Unify p(A, B) and p(x, x)

6. Unify p(x, f(x), B) and p(x, f(B), B)

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Quiz

7. Unify p(x, y) and p(y, A)

8. Unify p(f(y), x) and p(x, f(B))

9. Unify p(f(y), y, x) and p(x, f(A), f(z))

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Inference in FOPL

Resolution

- **Unification** ✓
- **Skolemization**
- **CNF**

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Conjunctive Normal Form (CNF)

The KB should be written in conjunctive normal form where each sentence of the KB is connect to another using conjunction.

EXAMPLE

$P \vee Q$	✓	$P \wedge Q$	✗
P	✓	$\forall x :$	✓
$\neg P$	✓	$\exists x:$	✗
$P \rightarrow Q$	✗		

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Convert to CNF

1- Eliminate all $\leftrightarrow$ connectives:

replace $(P \leftrightarrow Q)$ by $((P \rightarrow Q) \wedge (Q \rightarrow P))$

2- Eliminate all $\rightarrow$ connectives:

replace $(P \rightarrow Q)$ by $(\sim P \vee Q)$

3- Reduce scope of negation to a single predicate:

replace $\sim\sim P$ by P

replace $\sim(P \vee Q)$ by $(\sim P \wedge \sim Q)$

replace $\sim(P \wedge Q)$ by $(\sim P \vee \sim Q)$

replace $\sim(\forall x)P$ by $(\exists x)\sim P$

replace $\sim(\exists x)P$ by $(\forall x)\sim P$

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Convert to CNF

4- Standardize variables:

rename so each quantifier has its own unique variable name.

5- Eliminate existential quantification: **Skolemization**.

6- Remove universal quantification symbols.

7- Put in conjunctive normal form: Distribute $\vee$ into $\wedge$

- E.g., convert $(P \wedge Q) \vee R$ to $(P \vee R) \wedge (Q \vee R)$
- E.g., convert $(P \vee Q) \vee R$ to $(P \vee Q \vee R)$

8- Split each conjunct into a separate clause (a disjunction of literals)

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Skolemization

Objective

Want all variables universally quantified.

Skolem's Theorem

Every existentially quantified variable can be **replaced** by a unique **Skolem function whose arguments are all the universally quantified variables** on which the existential depends, without changing FOL.

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Skolemization

EXAMPLE

Skolemize the following expression:

$\forall x \exists y : person(x) \wedge thing(y) \rightarrow likes(x,y)$

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Skolemization

EXAMPLE

Skolemize the following expression:

$\forall x\ \forall y\ \exists z : p(x, y) \rightarrow q(y, z)$

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Skolemization

EXAMPLE

Skolemize the following expression:

$\forall x\ \exists y\ \forall z : p(x, y) \rightarrow q(y, z)$

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Skolemization (Quiz)

Skolemize the following expression:

$\exists x\ \forall y\ \forall z : p(x, y) \rightarrow q(y, z)$

Search For the Answer

Convert to CNF

EXAMPLE

Convert the following sentence to CNF:

$\forall x\ \forall y: p(x, y) \vee (q(x, y) \wedge r(x, y))$

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Convert to CNF

EXAMPLE

Convert the following sentence to CNF:

$\forall x \forall y\ \exists z : (above(x, y) \wedge \neg\ on(x,y)) \rightarrow (above(x, z) \wedge above(z, y))$

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Convert to CNF

EXAMPLE

Convert the following sentence to CNF:

$\forall x : (p(x) \vee q(x)) \rightarrow \exists y : w(x, y)$

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Resolution Steps in FOPC

1. Convert premises into CNF.

2. Add the negation of what to be proved to the set of premises.

3. Resolve the clauses together, producing new clauses that logically follow from them (Resolution Tree).

4. Produce a contradiction by generating an empty clause.

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Resolution in FOPC

EXAMPLE

- **Given (KB):**
  1. All dogs are animals.
  2. Fido is a dog.
  3. All animals will die.

- **Prove:**
  4. **Fido will die.**

KB |= Fido will die

Resolution in FOPC

1. Add the negation of the query to KB

KB

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Resolution in FOPC

2. Convert all sentences to FOPL

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Resolution in FOPC

3. Convert all sentences to CNF

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4. Apply Resolution Refutation Algorithm

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Resolution in FOPC

EXAMPLE

All people who are not poor and are smart are happy. Those people who read are smart. John can read and is not poor. Happy people have exciting lives.

Query:

Can anyone be found with an exciting life?

Resolution in FOPC

All people who are not poor and are smart are happy. Those people who read are smart. John can read and is not poor. Happy people have exciting lives.

Prove:

Can anyone be found with an exciting life?

Resolution in FOPC

Convert them to Conjunctive Normal Form.

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Resolution in FOPC

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Resolution in FOPC

EXAMPLE

Anyone passing his history exams and winning the lottery is happy. But anyone who studies or is lucky can pass all his exams. John did not study but he is lucky. Anyone who is lucky wins the lottery.

Prove:

John is happy.

KB |= John is happy

Resolution in FOPC

1. Add the negation of the query to KB

KB

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Resolution in FOPC

2. Convert all sentences to FOPL

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Resolution in FOPC

3. Convert to CNF

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4. Apply Resolution Refutation Algorithm

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Resolution in FOPC

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Resolution in FOPC

EXAMPLE

1. John likes all kind of food.
2. Apple and vegetable are food.
3. Anything anyone eats and not killed is food.
4. Anil eats peanuts and still alive.
5. Harry eats everything that Anil eats.

Prove by resolution that:

John likes peanuts.

Resolution in FOPC

2. Convert all sentences to FOPL

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Resolution in FOPC

3. Convert to CNF

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4. Apply Resolution Refutation Algorithm

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Resolution in FOPC

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Check - Chapter Learning Outcomes

CLO

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End of Chapter

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